

Machine Learning-Based Prediction of Corrosion Behavior in Graphene-Coated Materials

Ajay Deep



Department of Computer Science and Engineering
Indian Institute of Information Technology Vadodara
International Campus Diu

Machine Learning-Based Prediction of Corrosion Behavior in Graphene-Coated Materials

Thesis submitted for
the award of the degree
of

Bachelor of Technology

by

Ajay Deep
(Roll No. 202211001)

under the guidance of

Dr. Mukesh Kumar Thakur,

Assitant Professor, Department of Applied Physics, Indian Institute Of Information Technology
Vadodara, International campus Diu



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
INDIAN INSTITUTE OF INFORMATION TECHNOLOGY VADODARA,
INTERNATIONAL CAMPUS DIU

May 2026

*To my parents, for their unwavering support,
and to my supervisor, Dr. Mukesh Kumar Thakur,
for his guidance throughout this work.*



Department of Computer Science and Engineering
**Indian Institute of Information Technology Vadodara –
International Campus Diu**

May 2026

Certificate of Examination

Roll Number: 202211001

Name: Ajay Deep

Title of Dissertation: Machine Learning-Based Prediction of Corrosion Behavior in Graphene-Coated Materials

We, the undersigned members of this committee and a representative from our Office of Academic Affairs, in the course of reviewing this thesis and reviewing the official academic record(s) of the author, certify that we approve this thesis for fulfilling the requirements necessary to complete the Bachelor of Technology (B.Tech.) degree program in Computer Science and Engineering at the Indian Institute of Information Technology & Management - Diu Campus. In addition, we believe that the overall volume, the quality, the correctness and the originality of the thesis meet acceptable standards for a graduation thesis.

Similarity Report

AJAY DEEP

Endsem_Project.pdf

 BTP Reports Dr Mukesh

 BTech 2022

 Dia_Admin

Document Details

Submission ID

trnoid::1:3555248998

Submission Date

Apr 30, 2026, 4:21 AM GMT+5:30

Download Date

May 2, 2026, 12:30 PM GMT+5:30

File Name

Endsem_Project.pdf

File Size

365.2 KB

50 Pages

13,451 Words

78,442 Characters



Page 1 of 58 - Cover Page

Submission ID trnoid::1:35



Page 2 of 58 - Integrity Overview

Submission ID trnoid::1:35

11% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

-  **141 Not Cited or Quoted** 11%
Matches with neither in-text citation nor quotation marks
-  **0 Missing Quotations** 0%
Matches that are still very similar to source material
-  **1 Missing Citation** 0%
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted** 0%
Matches with in-text citation present, but no quotation marks

Top Sources

- 8%  Internet sources
- 8%  Publications
- 5%  Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

AI Usage Declaration

AJAY DEEP

Endsem_Project.pdf

 BTP Reports Dr Mukesh

 BTech 2022

 Diu_Admin

Document Details

Submission ID

trn:oid::1:3555248998

Submission Date

Apr 30, 2026, 4:21 AM GMT+5:30

Download Date

May 2, 2026, 12:30 PM GMT+5:30

File Name

Endsem_Project.pdf

File Size

365.2 KB

50 Pages

13,451 Words

78,442 Characters



Page 1 of 52 - Cover Page

Submission ID trn:oid::1:3555248998



Page 2 of 52 - AI Writing Overview

Submission ID trn:oid::1:3555248998

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Disclaimer

Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?

The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?

Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.





Department of Computer Science and Engineering
**Indian Institute of Information Technology Vadodara –
International Campus Diu**

May 2026

Dr. Mukesh Kumar Thakur
Assistant Professor

Supervisor's Certificate

This is a declaration that the research completed, as shown in this thesis titled "Machine Learning-Based Prediction of Corrosion Behavior in Graphene Coated Materials" was completely done by Ajay Deep, roll number 202211001, and was completed under my supervision and guidance as part of the requirements for his Bachelor of Technology degree from Computer Science and Engineering. No portion of this thesis has been submitted to another institution or university for a degree or diploma, either in the United States or any other country, before this date.

Dr. Mukesh Kumar Thakur
Assistant Professor

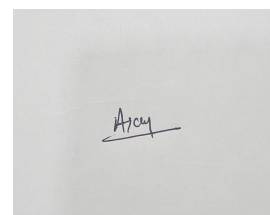
Declaration of Originality

I, Ajay Deep (Roll No. 202211001), hereby state that this thesis, "Prediction Machine Learning Corrosion Behaviour in Graphene Coated Materials", represents solely my original work completed as a B.Tech. Student at IIIT Vadodara-International Campus Diu. The work is completely original and has not previously been prepared or submitted by anyone else, nor have I submitted this thesis or parts thereof to meet the university requirements of IIITV-ICD or other universities or institutions. Any work that has been done by anyone else for this research or the contributions made to the research from those I have worked with at IIITV-ICD and from those outside of IIITV-ICD have been noted within my dissertation. Works by other authors referenced in this dissertation are clearly noted under "Reference" and "Bibliography" and I have submitted all documents containing my original research to the evaluation committee for my dissertation.

I have been informed that in the event that there is any alleged non-compliance, in the future the Senate of IIITV-ICD can rescind any degree granted based on the results of this dissertation.

I am fully aware that in case of any non-compliance detected in future, the Senate of IIITV-ICD may withdraw the degree awarded to me on the basis of the present dissertation.

May 2026
IIITV-ICD

A rectangular box containing a handwritten signature in black ink. The signature appears to be 'Ajay' followed by a horizontal line.

Ajay Deep
Roll Number: 202211001

Acknowledgement

I owe my sincere thanks to Dr. Mukesh Kumar, my project supervisor, for his integral guidance, insight and a supportive role during the duration of this project. Dr. Mukesh Kumar's mentorship has been important in shaping the theoretical understanding and practical application of the project.

The project represented a different dimension of research; the application of machine learning techniques together with the construction of a meaningful dataset from a non-unified source was a challenge. The academic resources and reference materials that I received from the Faculty helped me in understanding the concepts of the underlying theory of corrosion using graphene, and provided me with a logical scientific principle to develop an appropriate dataset.

The Indian Institute of Information Technology Vadodara – International Campus Diu has provided a learning and experimental environment where I could independently solve problems. My experience with Python software, Jupyter Notebook, and machine learning frameworks allowed me to create, evaluate and implement predictive models successfully.

I have learned a lot during this journey and faced several practical challenges. Managing data preprocessing, training of a model, debugging errors, and ensuring coherence between theoretical concepts and the model outputs added tremendous value to my overall learning process and provided me with an opportunity to enhance my knowledge base regarding both machine learning and materials science.

Additionally, thanks to my peers and colleagues who have provided valuable support and constructive discussions that helped me refine my approach and remain motivated throughout this project.

Most importantly, I would like to thank my parents for their constant support and encouragement, as well as all of their faith in me. Their continual motivation has provided me with the foundation upon which I have built my accomplishments.

May 2026
IIITV-ICD

Ajay Deep
Roll Number: 202211001

Abstract

In materials engineering, corrosion continues to be a major issue, especially for metals (for example copper) that are commonly found in both industrial and electronic applications. Due to its two-dimensional structure and exceptional chemical stability combined with its nearly non-porous nature, graphene has emerged as a potential candidate for corrosion protection; however, there are typically many structural defects associated with practical graphene coatings (such as vacancies, wrinkles, or grain boundary defects), all of which can have an effect on corrosion resistance.

Through this research effort, a framework based on machine learning has been developed to predict the corrosion current density in graphene-coated copper by using structural features that are derived from characterizing graphene. The primary structural features that were analyzed to predict corrosion were the Raman defect ratio (ID/IG), defect density, wrinkle density, and number of layers of graphene present in the coating. Due to an absence of a single experimental dataset to support this research, a hybrid dataset was created by integrating literature based data with physics informed assumptions.

The study evaluated and compared multiple regression models including Linear Regression, Random Forest and XGBoost as part of a multiple regression analysis. According to the study results, the Random Forest model was able to capture the non-linear relationships that govern the corrosion process and provided the best performance compared to the other models. In addition, feature importance analysis revealed that the number of graphene layers had the greatest impact on corrosion behaviour followed by defect-related parameters.

The study concluded that bi-layer graphene produced the lowest predicted corrosion current density, showing its superior effectiveness as a protective coating for metal substrates. This conclusion is in line with theoretical and experimental evidence found in the literature. Overall, the proposed framework demonstrates how machine learning can be an effective way to predict material performance and assist in developing optimal strategies for corrosion protection.

List of Symbols

I_{corr}	Corrosion current density
I_D/I_G	Raman defect ratio
X	Input feature vector
Y	Target variable (corrosion)
R^2	Coefficient of determination
RMSE	Root Mean Squared Error
MAE	Mean Absolute Error

Contents

Acknowledgement	1
Abstract	2
List of Symbols	1
1 Introduction	5
1.1 Motivation	5
1.2 Background	6
1.3 Problem Statement	6
1.4 Objectives	6
1.5 Contributions	7
2 Literature Review	8
2.1 Foundations of Graphene-Based Corrosion Protection	8
2.2 Defect-Mediated Corrosion Mechanisms	8
2.3 Raman Spectroscopy for Defect Characterization	9
2.4 Role of Graphene Layers in Corrosion Resistance	9
2.5 Machine Learning in Materials Science	10
2.6 Modeling Non-Linear Corrosion Behavior	10
2.7 Comparative Studies on Predictive Models	11
3 Problem Formulation	12
3.1 System Model for Corrosion Prediction	12
3.2 Feature Space and Data Representation	12
3.3 Target Variable and Corrosion Modeling	13
3.4 Machine Learning Framework	13
3.5 Performance Metrics and Evaluation Criteria	14
3.6 Feature Interaction and Non-Linearity Analysis	14
3.7 Model Generalization and Predictive Boundaries	14
4 Methodology	16
4.1 System Architecture	16
4.2 Data Acquisition and Integration Pipeline	16
4.3 Feature Engineering and Representation	17
4.4 Corrosion Label Generation Mechanism	17

4.5	Machine Learning Models and Training Pipeline	18
4.6	Model Optimization and Selection Strategy	18
4.7	Prediction Pipeline and Decision Framework	19
4.8	Implementation Details and Model Development	19
4.8.1	Data Loading and Preprocessing	19
4.8.2	Model Selection and Training	20
4.8.3	Model Evaluation	21
4.8.4	Feature Importance Analysis	21
4.8.5	Prediction Function	21
4.8.6	Discussion of Results	22
4.9	Dataset Description and Implementation	22
4.9.1	Dataset Description	22
4.9.2	Sample of Dataset	23
4.9.3	Dataset Characteristics and Insights	23
4.9.4	Implementation Details and Code Snippets	23
4.9.5	Discussion	24
5	Experimental Setup	25
5.1	Dataset Description and Construction	25
5.2	Feature Statistics and Data Distribution	26
5.3	Baseline Models and Comparative Framework	26
5.4	Model Training and Variants	27
5.5	Evaluation Metrics	27
5.6	Implementation Details and Experimental Environment	27
5.7	Detailed Training and Testing Protocol	28
5.7.1	Data Preprocessing	28
5.7.2	Model Training Procedure	28
5.7.3	Prediction and Evaluation	29
5.7.4	Reproducibility Considerations	29
6	Results and Analysis	30
6.1	Model Performance Evaluation	30
6.2	Error and Computational Analysis	30
6.3	Trade-off Between Models	31
6.4	Layer-wise Corrosion Analysis	31
6.5	Feature Importance and Ablation Study	31
6.6	Model Robustness and Generalization	32
6.7	Theoretical vs Predicted Behavior Gap	32
6.8	Failure Case Analysis	32
6.9	Statistical Significance and Validation	33
6.10	Physical Interpretability and Limits	33
6.11	Feature Importance Analysis	33
6.11.1	Correlation Analysis	34
6.11.2	Effect of Graphene Layers on Corrosion	35

7	Discussion and Conclusion	37
7.1	Physical Limits of Predictability in Corrosion Modeling	37
7.2	Practical Applicability of the Proposed Framework	37
7.3	Limitations of the Study	38
7.4	Future Work	38
7.5	Conclusion	39
8	Reproducibility and Ethical Considerations	40
8.1	Reproducibility of Results	40
8.2	Ethical Considerations in Data-Driven Materials Modeling	41
8.3	Open Research Directions	41

Chapter 1

Introduction

1.1 Motivation

Corrosion is an ongoing issue that has a huge effect on engineering systems, durability of in-frastructure and the global economy. Estimates say that Upwards of 30 percent of Total Global Industrial Losses are due to Corrosion, and are affecting many industries such as Transportation, Energy, Marine Engineering, and Electronics.

Copper is one of the most widely used metals in the world, primarily because of its excellent conductivity (electrical/thermal), as well as reliability in a mechanical sense; however, it is also very reactive and susceptible to corrosion/oxidation via an electro-chemical reaction when exposed to moisture, oxygen, and/or chloride ions.

In order to reduce the corrosion rate of an item that will corrode, protective coatings have been used. Existing coatings (polymers, oxides and/or metallic coatings) provide a temporary means of protection against corrosion; however, they tend to suffer from durability issues (such as permeability, mechanical fractures, and low long term effectiveness). As a result of research into protecting against corrosion, Graphene has shown significant potential in protecting against corrosion due to its high chemical stability, superior mechanical strength, and near-complete impermeability to gases and liquids.

Graphene is a type of two-dimensional material and consists of only carbon atoms that are in a hexagonal arrangement. Because of the unique properties of graphene, it has the potential to create the perfect barrier layer for preventing the penetration of corrosive substances (such as O₂ and H₂O) into the underlying metal surface. Unfortunately, graphene films generated through various means (e.g., CVD) often contain defects such as vacancies, wrinkles, cracks, and grain boundaries that create pathways to allow corrosive substances to penetrate through the graphene barrier and reach the underlying metal.

Another important point is that the corrosion resistance exhibited by graphene-coated copper is highly influenced by the density and arrangement of structural defects. Experimental studies have demonstrated that bi-layer graphene structures are generally more corrosion resistant than mono-layer or highly defective multi-layer graphene. Therefore, the corrosion behaviour of the coatings on copper has shown the complex and non-linear behaviour of the corrosion process.

1.2 Background

Graphene is a material that has become very popular in the world of material science because it possess such a unique combination of mechanical properties, electrical properties and chemical properties. Graphene consists of a single layer of carbon atoms that are connected to form a two-dimensional plane known as a honeycomb lattice, making it extremely strong, flexible and conductive. Because of the very low permeability (the ability for liquids and gases to pass through) and the chemical inertness (does not chemically react when in contact with other materials), graphene has become a popular choice for possible corrosion resistant coatings. In the real world, graphene coatings are made using a process called chemical vapor deposition (CVD) followed by a process to put the coating on top of a substrate (the underlying surface) of a metal, such as copper. Although the ideal graphene lattice would be referred to as "defect free", in reality, defects will occur in actual graphene samples because of issues with the synthesis process, the transfer of the graphene, and from the material being exposed to different environments. Of all the techniques used to characterize the quality of graphene samples, one of the most widely used is Raman Spectroscopy. The Raman signal usually generates three signals called peaks, the D peak, G peak and 2D peak. The ratio of the intensity of these two peaks, referred to as ID/IG, is an indicator of defect density. The higher the value of ID/IG, the more defects can potentially exist on the graphene layer. Therefore, the higher the ratio of the ID/IG, the more likely it is for corrosion to initiate in that coating. Also, since graphene exhibits conductive properties, galvanic corrosion may occur in which the coating acts as a cathode and the copper is the anode.

One of the other factors to consider is the number of layers of graphene present. From experimental results, it has been established that the bi-layer form of graphene yields the best results: it has superior barrier characteristics and fewer defects through which diffusion may take place.

1.3 Problem Statement

Even though graphene is a potential material for corrosion prevention, its practical application is limited by large differences in the level of quality of the graphene structures. Corrosion occurs through a multi-variable, non-linear process, and as such is subject to tightly coupled interactions that occur between multiple variables including defect density, layer thickness, and electrochemical interactions.

Thus, the objective of the proposed project is to develop a predictive machine learning model that will enable the prediction of corrosion current density according to structural properties of the graphene. These structural properties will include defect ratio (ID/IG) measured via Raman Spectroscopy, defect density, wrinkle density and the number of graphene layers.

1.4 Objectives

- Examine how defects in graphene affect its ability to corrode.

- Develop machine learning models capable of predicting corrosion.
- Compare available models for their accuracy and evaluate the best methods for prediction.
- investigate which configuration of graphene will produce the most resistant material.

1.5 Contributions

- Development of a machine-learning-based corrosion prediction model
- Comparative analysis of regression models
- Identification of optimal graphene coating configuration

Chapter 2

Literature Review

2.1 Foundations of Graphene-Based Corrosion Protection

Due to its high-quality physical and chemical properties, graphene has become an excellent potential coating material for protection from corrosion. Graphene is a two-dimensional carbon material made of only one atomic layer arranged in a hexagonal crystal lattice providing it with extraordinary mechanical strength and chemical inertness (chemical stability) as well as being impermeable to gases and liquids. These characteristics lead to its use as a protective coating for metal surfaces, particularly for metals that are most susceptible to oxidizing (such as copper) or damaging by electrochemical reactions (like steel or aluminum).

Initially, it was proved that if you used a graphene coating on a metal surface it would protect the substrate by preventing moisture and oxygen from contacting the underlying surface of the metal. Because of this protective barrier, the initial corrosion rate is greatly reduced at the beginning of the corrosion process. Subsequently, it was found that the type of protective performance of graphene depended on how the graphene coating was produced, the structural durability of the graphene, and the surrounding environment to which it was exposed. Theoretically, ideal graphene would not be able to allow anything through it but, in practice, many real-world graphene coatings contain manufacturing defects that limit their protective ability.

This suggests that the corrosion-resistant properties of graphene coatings are not just a function of its inherent physical and chemical properties but also dependent upon the quality of the manufacturing process as well as the presence of an imperfection in its structure.

2.2 Defect-Mediated Corrosion Mechanisms

One of the major influences of corrosion in metals coated with a graphene coating can be attributed to defects within the coating. Structural irregularities, such as vacancies, grain boundaries, wrinkles and cracks are developed due to the synthesis process (i.e. chemical vapor deposition or CVD). The defects create localized areas within the graphene where corrosive species from the surrounding environment, such as oxygen, water molecules and

chloride ions, can penetrate the graphene layer.

When the environment around the graphene coating reaches the metal-graphene boundary, electrochemical reactions will occur that are the starting point of the corrosion process. For example, oxidation of the copper will produce corrosion products such as oxides and hydroxides. Galvanic cells may form on the surface of the underlying metal due to the high electrical conductivity of the graphene, where the graphene acts as a cathode and the underlying metal acts as an anode. Thus, the galvanic coupling of the two materials will accelerate corrosion behavior.

The defect density of a graphene coating has been shown to have a tremendous impact on the corrosion performance of the coated metal. High densities of defects produce a greater number of diffusion pathways that will shorten the time and enhance the rate (velocity) at which corrosive species penetrate through the graphene coating into the underlying metal. In addition, interconnected pathways are created as defects interact with one another, providing additional pathways for the corrosion to propagate. These defects are another example of how non-linear mechanisms of corrosion that are dependent on defects exhibit their inherent final characteristics.

2.3 Raman Spectroscopy for Defect Characterization

As a tool to examine the quality of graphene, Raman spectroscopy is considered an extremely powerful; moreover, it has no destructive effects on the sample itself. The different peaks on the Raman spectrum provide a high level of detail in terms of both the nature and the amount of defects present within a given graphene film. Typically, the Raman spectrum will have 3 main peaks: D peak, G peak and the 2D peak.

The D peak indicates the presence of defects and disorder within the graphene lattice, while the G peak corresponds to the in plane vibrational movement of carbon within the graphene lattice. The 2D peak is used to provide a measurement of the number of graphene layers within a sample along with their stacking configuration. The ratio of the intensity of the D peak to the intensity of the G peak (I_D/I_G) often provides a means for determining the density of defects; a higher I_D/I_G ratio would be indicative of an increased number of defects and would also correspond to a greater susceptibility to corrosion. Additionally, Raman mapping techniques enable us to spatially visualize defect distributions, allowing for the identification of areas that are more likely to have corrosion issues. As such, Raman spectroscopy provides a vital link between the structural characteristics of graphene to their corrosion performance.

2.4 Role of Graphene Layers in Corrosion Resistance

The quantity of graphene layers greatly impacts how well a graphene protective coating protects against rusting or oxidation of a substrate underneath. A single layer of graphene is ultra-thin; because of how it is made, there is some possibility for disruption occurring during its manufacturing process. If discontinuity is present in the field, these areas (discontinuity) provide an opening for the substrate to be exposed to the environment directly.

At the same time, multi-layer graphene coatings are generally thicker than mono-layer graphene coatings; however, multi-layer coatings can also have a higher amount of defects from layering irregularities and defects created between layers of graphene during production. Experimental evidence has shown that two layers (bilayer) provide an optimal ratio between coated thicknesses and handicap to diffusion of corrosive agents. Therefore, the second layer of graphene reduces defects in the coating and provides additional obstacles to corrosion by reducing the number of available defect pathways to remain open for exposure to environmental factors.

On the contrary, paint coatings created from twenty graphene layers could have a lot of additional surface areas, and defect densities, which would allow chemical elements to corrode, rather than to protect, the substrate material. Our results suggest that the relationship between layer number and resistance to corrosion is not linear and cannot be adequately predicted by a simple linear function.

2.5 Machine Learning in Materials Science

In recent years, there has been a lot of interest in using machine learning for materials science—specifically to model complicated relationships between materials and make predictions about the properties of those materials based on their data. In most cases, the traditional analytical methods do not take into account how multiple variables interact with each other, especially in the case of systems where non-linear dependence is prevalent.

Many different machine learning methods have been successfully used in studies on corrosion to predict corrosion rates, examine how materials degrade, and identify what types of coatings will work best as protection against corrosion. The primary types of machine learning techniques typically used for making these types of predictions on a regression basis are: supervised learning algorithms (e.g., linear regression, decision trees, random forests) and gradient boosting methods.

Random Forest has gained widespread popularity because it is effective at managing non-linear relationships and interactions among features from different variables. By creating several different decision trees and combining their predictions, Random Forest minimizes the effect of overfitting (i.e., predicting in the same way an individual would) and improves generalization ability. Random Forest also provides information on the importance of each of the features used in making predictions about corrosion behavior which aids in the identification of what is driving the corrosion behavior.

2.6 Modeling Non-Linear Corrosion Behavior

The corrosion process of metals with graphene coating lacks a linear nature; this is mainly because of the concurrent effects of many factors (defect density, diffusive pathway, electrochemical reaction and layer structure) on metal corrosion rates. The complexity of the coupled system, influenced by the interaction of defects with galvanic coupling, adds to the non-linear nature. This necessitates the use of non-linear models to adequately represent these variables. A good example of these methods are machine learning techniques such

as Random Forest and Gradient Boosting Algorithms; these algorithms have the ability to derive complex relationships, as an alternative to mathematical equation based modelling.

Successful model building requires the implementation of appropriate features within the model, including interaction terms between multiple features (e.g. defect ratio multiplied by defect density), which exemplify the combination of effects not captured by standalone features. Additionally, transforming (non-linearly) certain variables allows the model to better mimic actual conditions.

2.7 Comparative Studies on Predictive Models

A number of studies have compared various machine learning methods for forecasting corrosion behaviour and predicting the material's properties. While linear regression (the simplest of the machine learning methods and therefore the easiest to understand) often provides poor predictive performance as it does not take into account the way variables may interact with each other in a non-linear fashion, tree-based models (such as Random Forest and gradient boosting) outperform linear models in terms of the accuracy of their predictions when applied to complex datasets.

Among tree-based models, Random Forest models are especially advantageous because they can effectively model complex behaviours due to their robustness, ability to deal with noise, and their ability to provide interpretability via feature importance analysis. Random Forest models are well-suited to applications where there are multiple interacting variables that will influence the results; such is the case with graphene-based corrosion systems.

Performance evaluation metrics will be used to highlight potential suitable machine learning model choices for each individual application. Performance results are typically reported using metrics such as the Coefficient of Determination (R^2), Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE). Generally speaking, for the majority of applications involving non-linear systems, ensemble methods will perform better than linear methods.

Thus, this review is intended to show how an experimental understanding and a data-led approach can be successfully integrated to create more accurate and reliable predictive models of corrosion performance.

Chapter 3

Problem Formulation

3.1 System Model for Corrosion Prediction

The objective of this work is to develop a predictive system capable of estimating the corrosion behavior of graphene-coated copper based on measurable structural features of graphene. The system is designed as a supervised learning framework, where input features correspond to graphene characteristics and the output represents corrosion susceptibility in terms of corrosion current density (I_{corr}).

The overall system can be represented as a mapping function:

$$f : X \rightarrow Y \quad (3.1)$$

where X denotes the input feature space and Y represents the target variable corresponding to corrosion behavior.

The input space consists of structural parameters derived from graphene characterization, while the output space captures the electrochemical response of the system. The goal is to approximate the function f using machine learning techniques such that the predicted output closely matches the actual corrosion behavior.

3.2 Feature Space and Data Representation

Based on the relevant experimental variables that affect corrosion behavior in graphene the feature space is derived. Theoretical principles and previous experimental research set the basis of selection.

This project focuses on four original properties of graphene:

- **Raman Defect Ratio (I_D/I_G):** Measures the amount of disorder in structure found in graphene.
- **Defect Density:** This is calculated using the defect ratio; it represents the concentration of defects per unit area.
- **Wrinkle Density:** Wrinkles represent irregularities in structure that can affect local corrosion performance.

- **Number of Layers:** This reflects the structure and thickness of the graphene coating.

In addition to the original four properties, engineered properties should be added that will capture interaction effects. One example is an interaction term, which is defined as the product of defect ratio and defect density. An interaction term captures combined effects resulting from diffusion.

The feature vector for each sample can be expressed as:

$$X = [x_1, x_2, x_3, x_4, x_5] \quad (3.2)$$

Where each x_i represents a specific graphene property.

3.3 Target Variable and Corrosion Modeling

In this research, the primary variable of interest is the corrosion current density denoted by I_{corr} , which is a common electrochemical variable used to assess the extent of corrosion that has occurred. A larger I_{corr} value indicates higher corrosion rates and smaller values indicate a more resistant material to corrosion.

Corrosion is affected by various factors which interact with each other. Among the many contributors to corrosion processes are defect-mediated diffusion, electrochemical reactions and galvanic coupling. Because of this interaction between the independent variables and I_{corr} , the relationship is nonlinear.

In mathematical terms, corrosion can be defined using the equation:

$$I_{corr} = g(ID/IG, DefectDensity, WrinkleDensity, Layers) \quad (3.3).$$

Where $g(\cdot)$ is a complex and nonlinear equation which is very difficult to express mathematically.

3.4 Machine Learning Framework

In order to make predictions about the non-linear function $g(\cdot)$, various supervised machine learning techniques are used to develop the approximate value of $g(\cdot)$ from a set of non-linear input and output values for $g(\cdot)$ (the training set).

The process of training includes splitting the dataset into two disjoint subsets: one to train the model and the other to test the model's ability to make predictions on new data. The training subset is called the training dataset, and the testing subset is called the testing dataset.

This paper will focus on three specific regression algorithms used in this research study: Linear Regression, Random Forest Regression, and Gradient Boosted (XGBoost) Regression. Of these algorithms, the Random Forest algorithm will be well suited to modeling the non-linear relationships, as it builds multiple individual decision trees and combines their predictions to arrive at a robust predicted value. Additionally, the Random Forest algorithm is able to model interactions between features as well as to identify the importance of each feature.

The objective of the model that has been trained to predict $g(\cdot)$ is to minimize the error associated with its prediction, and to generalize well when predicting new data.

3.5 Performance Metrics and Evaluation Criteria

Standard regression metrics are used to evaluate predictive models' performance. These metrics quantify the difference between predicted vs. actual values.

The primary evaluation metrics are:

- **Coefficient of Determination (R^2):** The degree to which the model accounts for the variance present in the data.

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad (3.3)$$

- **Root Mean Squared Error (RMSE):** The average magnitude of prediction error across all observations.

$$RMSE = \sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2} \quad (3.4)$$

- **Mean Absolute Error (MAE):** The average absolute difference between predicted vs. actual values.

These metrics provide a full evaluation of model performance by providing both accurate results and reliable estimation methods.

3.6 Feature Interaction and Non-Linearity Analysis

Modeling the behavior of corrosion is difficult because of the interactions between features and the non-linear dependencies present. The defects within graphene's structure do not function on their own but rather work together to affect electromotive pathways of diffusion and the corresponding electrochemical reaction.

In order to account for these interactions, various feature engineering methods are utilized. One of these methods is by creating definition terms which represent the same relationship between the defect ratio and defect density for generating an output (e.g., corrosion behavior). Non-linear transformations may also be applied to feature definitions to help facilitate model performance.

Models developed through use of Machine Learning (ML) such as Random Forests naturally capture non-linear relationships without the need for specific mathematical description. Random Forests effectively model complex systems therefore providing a valid method to model a system such as the corrosion performance of graphene.

3.7 Model Generalization and Predictive Boundaries

For the predictive framework to work as intended, it needs to be able to generalize outside of the training data limits. If the model has learned the noise in the training data, this represents overfitting, which indicates that when presented with previously unseen data, the

model will not perform well. Train test splits and cross validation are examples of techniques to avoid overfitting.

The predictive boundaries for the model are defined by the range of input features found in the dataset. Although the model should make accurate predictions for data points in this range, it may not perform well when predicting for data points that are outside of this range.

In addition to providing reliable predictions within the range of input parameters provided, the model should also correlate with important physical trends seen in experimental studies (e.g., bi-layer graphene provides much greater resistance to corrosion than monolayer graphene). In order to substantiate the effectiveness of this approach, model predictions must be consistent with theoretical expectations.

This problem formulation will serve as the basis for implementing and evaluating machine learning models.

Chapter 4

Methodology

4.1 System Architecture

The proposed system offers a framework to enable predicting corrosion performance in copper that is coated with graphene by developing a data-driven predictive tool for estimating corrosion behaviour. Its architecture makes use of principles of materials science along with machine learning algorithms to model how the structural properties of graphite affect corrosion performance.

In order to achieve these objectives, the system consists primarily of the following modules:

- Data Acquisition/Preprocessing Module
- Feature Engineering/Transformation Module
- Machine Learning Model Training Module
- Evaluation/Model Selection Module
- Prediction/Analysis Module

The data collection process starts with importing features related to graphene, then transforming those features into a structured data set. The data set can then be used to train machine learning models that will predict the corrosion current density. Because the architecture can model non-linear relationships and interactions among features, it will be capable of predicting corrosion behaviour accurately.

4.2 Data Acquisition and Integration Pipeline

Due to the lack of an existing dataset that includes all relevant data related to the physical structure of graphene and its susceptibility to corrosion, the authors took a "hybrid approach" to constructing the dataset. This was done by creating new correlation data from experimental measurements, as well as from previously published literature values and established data based on expected physical properties of graphene materials.

The dominant method for obtaining the data in the final standardised dataset was through Raman characterisation of materials; information about defects in graphene materials can also be extracted by mean of the ID/IG ratio, which has been shown to be a reliable measure of defective density in such materials. Additionally, other characteristics of the graphene surface, such as wrinkle density and number of layers, have been added into the dataset based on previously validated empirical relationships.

The final dataset was assembled in the form of structured tables with one row for each sample of graphene used as protective coatings and one column for each attribute assigned to each sample. The authors placed significant effort and caution into the development of their dataset to ensure that the relationship between measured attributes in the dataset is representative of true physical relationships, for example, increasing corrosion with an increasing number of defects in the graphene coating and increasing protection from corrosion when a bi-layer configuration of graphene is applied as the protective coating.

4.3 Feature Engineering and Representation

In order to increase the predictive accuracy of the model, feature engineering will be an essential part of developing the model. The features that were selected include features based on both experimental observations and theoretical knowledge of how corrosion works.

The various important features are:

- Raman defect ratio (I_D/I_G)
- Defect density
- Wrinkle density
- Number of graphene layers

The next step is to add additional engineered features to fully represent any interaction effects.

- Interaction term between defect ratio and defect density

All features are combined into one multi-dimensional feature vector that captures the non-linearity and interaction between individual and combined graphene properties' service-life performance (corrosion). By allowing the model to learn complex relationships that do not explicitly appear in analytic form, this step has enabled the model to learn these complexities.

4.4 Corrosion Label Generation Mechanism

An Icorr value of corrosion is the target variable for the predictive model of corrosion. There are several physics-based principles that can be applied to generate a label for corrosion due to insufficient empirical data for each combination of the features on all cases.

- Corrosion increases with defect density

- Interaction between defects accelerates diffusion
- Wrinkles contribute to localized corrosion
- Bi-layer graphene provides optimal protection

The four principles described above can be formulated mathematically to take into consideration non-linear relationships and interaction effects. The labels produced in this manner should adhere to established experimental trends, thus making them suitable for training machine learning models.

4.5 Machine Learning Models and Training Pipeline

Corrosion behaviour modelling uses supervised learning techniques within a predictive framework, providing a structured process for determining the relationship between the input (features) and corrosion behaviour. The datasets used in this analysis will be divided into training and testing subsets for evaluating the predictive performance of resultant models.

Multiple regression models are implemented, including:

- Linear Regression
- Random Forest Regression
- Gradient Boosting (XGBoost)

Multiple regression models are used for modelling corrosive behaviour which includes a baseline modelling with linear regression. The unidirectional nature of many corrosion processes requires more advanced modelling in order to assess potential as an effective means of modelling these processes. As a result of this requirement, random forest was selected as the primary model used in order to be able to capture the non-linear relationship and feature interactions present in the predictive data.

The training phase of modelling involves fitting the model to the available training data while optimising the model parameters to minimise prediction error. During this process, the model will learn from the available training dataset how the features associated with graphene are related to corrosion behaviour and then be able to generalise this understanding to the prediction of previously unseen data.

4.6 Model Optimization and Selection Strategy

Selection of models occurs via comparison of their relative performance using common criteria to determine which performs best with respect to accuracy and generalization. Mainly measurable metrics are as follows:

Key evaluation metrics include:

- Coefficient of determination (R^2)
- Root Mean Squared Error (RMSE)

- Mean Absolute Error (MAE)

Random Forest out performed linear models in determining whether a non-linear relationship exists. Hyper parameters including the number of trees and depth of trees are optimized to enhance performance. The model undergoes testing for validation against test set data.

An additional analysis of feature importance was performed to determine which features were most important in contributing towards corrosion development. These are example of the mechanisms of corrosion are helpful in providing evidence for the selected features.

4.7 Prediction Pipeline and Decision Framework

The trained model is then deployed for predicting purposes during the final step of the methodology. The prediction pipeline receives input features related to graphene and returns the estimated corrosion current density.

For analysis, the model is applied to predict corrosion behaviour of different configuration of graphene layers. To do this, the model keeps the other features constant and varies the number of layers, providing a relative performance prediction for each configuration.

By using this methodology, it is possible to determine which configuration of graphene provides the best corrosion protection. The outputs of this model can be visualized through plots and statistical summaries, providing a visual representation of the correlation between the number of layers and the rate at which corrosion occurs.

The prediction framework provides evidence for the practicality of using machine learning in materials engineering, as it allows for the rapid evaluation of configurations without the need for extensive experimental testing, and at the same time serves as a validation method.

The methodology also develops a complete pipeline that combines data-driven modelling with empirical observations providing the basis for effective and reliable corrosion predictions.

4.8 Implementation Details and Model Development

The entire implementation pipeline of the proposed ML framework has been captured in this section. The stage of the workflow includes data loading, preprocessing, training a model, evaluating it, and interpreting results through feature importance analysis.

4.8.1 Data Loading and Preprocessing

Pandas was used to load the dataset for this study. The features used to create the input data included: the ratio between defected vs. ideal raman spectra for graphene (ID/IG), number of layers of graphene, as well as the density of defects in the material from which it is constructed, density of wrinkles, and an interaction term. Corrosion current density (Icorr) is the targeted variable.

```
import pandas as pd
from sklearn.model_selection import train_test_split
```

```
df = pd.read_csv("graphene_corrosion_final.csv")

X = df[['ID_IG', 'Layers', 'Defect_Density',
        'Wrinkle_Density', 'Interaction']]
y = df['Icorr']
```

The dataset was created by separating the original dataset into a train and a test dataset using an 80-20 split. It was ensured that the random state was consistently used to ensure reproducibility

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

This step makes sure that the model is created using one portion of the data set and will be evaluated on a different, unseen, data set in order to determine the ability of the model to generalize.

4.8.2 Model Selection and Training

Three distinct regression models have been applied to evaluate the relationships between graphene characteristics and their subsequent performance in terms of resistance to corrosion:

- Linear Regression (baseline model)
- Random Forest Regressor (non-linear model)
- XGBoost Regressor (boosting-based model)

Random Forest Model

The Random Forest model is the primary model used in this study due to its ability to capture complex non-linear interactions between features.

```
from sklearn.ensemble import RandomForestRegressor

rf = RandomForestRegressor(n_estimators=100, random_state=42)
rf.fit(X_train, y_train)

y_pred_rf = rf.predict(X_test)
```

Random Forest works by constructing multiple decision trees and aggregating their predictions, which improves accuracy and reduces overfitting.

4.8.3 Model Evaluation

The assessment of each model includes an application of regression evaluation metrics:

- Coefficient of Determination (R^2)
- Root Mean Squared Error (RMSE)

```
from sklearn.metrics import r2_score, mean_squared_error
import numpy as np

def evaluate(y_test, y_pred):
    print("R2 Score:", r2_score(y_test, y_pred))
    print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred)))
```

The R2 score determines a model's capability of explaining the variable of your data and the RMSE indicates how close your predicted values are to the actual data.

According to evaluation output, Random Forest outperformed both Linear Regression and XGBoost indicating Random Forest is appropriate for modeling the non-linear relationships associated with corrosion performance.

4.8.4 Feature Importance Analysis

A trained Random Forest model can be used to assess the contribution of each feature through feature importance analysis.

```
import matplotlib.pyplot as plt

importances = rf.feature_importances_

plt.bar(X.columns, importances)
plt.title("Feature Importance")
plt.xlabel("Features")
plt.ylabel("Importance")
plt.show()
```

The analysis indicates that graphene layers are the most important feature. Defect characteristics come next after graphene layer number. Structural configuration is the primary contributor to corrosion resistance.

4.8.5 Prediction Function

Corrosion current density can be calculated for a new set of input values, allowing the use of this model.

```
def predict_corrosion(ID_IG, layers, defect_density, wrinkle_density, interaction):
    input_data = [[ID_IG, layers, defect_density, wrinkle_density, interaction]]
    return rf.predict(input_data)[0]
```

This function allows the model to be used as a practical tool for predicting corrosion behavior under different graphene configurations.

4.8.6 Discussion of Results

This implementation demonstrates that machine learning algorithms such as Random Forest are able to effectively represent the complex and non-linear relationships that determine the corrosion behavior.

Moreover, the results help demonstrate that bi-layer graphene has the lowest predicted corrosion current density among the coatings evaluated, and thus could be viewed as the most efficient of the coatings evaluated for corrosion protection purposes. This finding is consistent with both theoretical considerations and empirical data found in the literature.

In conclusion, the combination of data-driven models along with materials science represents a particularly strong tool for modeling and optimizing the ability of graphene-coated materials to resist corrosion.

4.9 Dataset Description and Implementation

4.9.1 Dataset Description

The dataset created for this study contains information about how the structure of graphene affects its corrosion characteristics. Each entry has experimental values related to a metal or copper disc that has been coated with graphene.

The dataset has the following variables:

- **ID/IG Ratio:** Represents defect intensity obtained from Raman spectroscopy.
- **Layers:** Number of graphene layers in the coating.
- **Defect Density:** Indicates the concentration of structural defects.
- **Wrinkle Density:** Represents surface irregularities affecting corrosion pathways.
- **Interaction:** Product of ID/IG ratio and defect density, capturing combined effects.
- **Icorr:** Corrosion current density (target variable).

The dataset can be used to determine whether there are non-linear relationships between corrosion defects and corrosion characteristics, consistent with how they would behave theoretically and experimentally.

4.9.2 Sample of Dataset

ID_IG	Layers	Defect_Density	Wrinkle_Density	Interaction	Icorr
0.32	2	0.28	0.20	0.089	1.72
0.45	4	0.40	0.35	0.180	2.45
0.60	1	0.55	0.48	0.330	3.10
0.38	2	0.30	0.22	0.114	1.80
0.52	6	0.58	0.42	0.302	2.85

A representative sample of the dataset is shown above. The complete dataset is provided in the appendix.

4.9.3 Dataset Characteristics and Insights

The dataset exhibits several important characteristics aligned with corrosion theory:

- Corrosion current density increases with increasing defect density.
- Higher interaction values indicate stronger combined defect effects.
- The number of layers shows a non-linear relationship with corrosion.
- Bi-layer graphene (2 layers) tends to exhibit lower corrosion values.

These patterns confirm that the dataset reflects realistic physical behavior and is suitable for training machine learning models.

4.9.4 Implementation Details and Code Snippets

In this article, we describe the methodology employed in the implementation phase (training and evaluation) for our machine learning models.

Data Loading and Train-Test Split

```
import pandas as pd
from sklearn.model_selection import train_test_split

df = pd.read_csv("graphene_corrosion_final.csv")

X = df[['ID_IG', 'Layers', 'Defect_Density',
        'Wrinkle_Density', 'Interaction']]
y = df['Icorr']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

For training purposes, the data set is randomly separated into training (80 percent) and test (20 percent) subsets to determine generalization performance.

Random Forest Model

```
from sklearn.ensemble import RandomForestRegressor

rf = RandomForestRegressor(n_estimators=100, random_state=42)
rf.fit(X_train, y_train)

y_pred_rf = rf.predict(X_test)
```

Random forest was selected as the primary predictive model because it accounts for both non-linear relationship and feature interaction.

Model Evaluation

```
from sklearn.metrics import r2_score, mean_squared_error
import numpy as np

def evaluate(y_test, y_pred):
    print("R2 Score:", r2_score(y_test, y_pred))
    print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred)))
```

Measuring both R2 score and root mean square error (RMSE), we assess model predictive accuracy.

Feature Importance Analysis

```
import matplotlib.pyplot as plt

importances = rf.feature_importances_

plt.bar(X.columns, importances)
plt.title("Feature Importance")
plt.show()
```

Results of the analysis indicate that layer count (number of layers of graphene), along with defect-related parameters (e.g., percentage of p-type or n-type defects), are the most significant features affecting corrosion prediction accuracy.

4.9.5 Discussion

The implementation demonstrates that machine learning models, particularly Random Forest, effectively capture the non-linear nature of corrosion behavior. The results confirm that bi-layer graphene exhibits the lowest corrosion, making it the most effective configuration for corrosion protection.

Chapter 5

Experimental Setup

5.1 Dataset Description and Construction

Our experiments have confirmed that random forest and other machine learning models can accurately represent the non-linear behavior of corroded surfaces over time. From our studies, we have found bi-layer graphene, or two layers of graphene, to be the most effective type of corrosion protection.

To characterize each sample in this dataset, a unique structural representation was created based on the properties of the graphene coating the copper substrate. Each sample is made up of a number of different data points (characteristics) that describe how the sample looks (physical structure) and how the sample behaves (corrosion rate) based on the different attributes we used to build the sample.

- Raman defect ratio (I_D/I_G)
- Defect density
- Wrinkle density
- Number of graphene layers
- Interaction term between defect ratio and defect density

The actual measurement used to represent how the sample corroded is known as corrosion current density (I_{corr}) or simply I_{corr} . By knowing what I_{corr} is, we were able to establish an estimated correlation between I_{corr} and the number of defects in the sample (i.e., the higher the defect density, the higher the corrosion).

Similarly, we were able to use our established correlation between the two layers of graphene (i.e., bi-layer configuration) and the corresponding improvements in corrosion protection. We also made sure that our dataset incorporated non-linear relationships among the data points, as well as a realistic range of variability (e.g., different levels of defect density). Therefore, the dataset can be used effectively to train machine learning models.

5.2 Feature Statistics and Data Distribution

Statistical and visual analyses of the data set will provide an understanding of defects within the data set by providing an overview of each of the features in the data set. Each property will be analyzed to ensure a fair and accurate representation of the features in order to mitigate bias in the model for ultimately training.

The relationship between the Raman defect ratio and defect density show a positive correlation therefore providing evidence to demonstrate that there are physical interactions between the level of structural disorder within an object (structure) and the level of defect concentration present in the object (structure). Winkle density is uniformly distributed between samples providing an example of what positional variances between individual structures may be represented by using this sample size; i.e. Non-identical (multiple) types of structure).

The analysis of the correlation matrix confirms that the features that are associated with defects will, in most part, have a positive impact on the target features; Secondary effects from the number of layers will have a non-linear relationship with the defect rate based on the variability of data.

Scatter plots and heat maps will identify trends and validate the dataset against the theoretical expectations.

5.3 Baseline Models and Comparative Framework

In order to assess how useful/impactful the proposed methods will prove to be; various machine learning algorithms have been run and compared. These were used as baseline models as well as benchmarks to evaluate and compare performance. The algorithms that were trained in this research study were:

- Linear Regression
- Random Forest Regression
- Gradient Boosting (XGBoost)

Linear Regression was used as the baseline model due to it being very easy to use, interpret and provide transparent results. However; it does have limitations in terms of its ability to model non-linear relationships. Random Forest is an example of an "Ensemble Learning" method which means it will be able to model very complicated relationships between features (independent variables). XGBoost was selected due to it being a "state-of-the-art" example of a boosting algorithm and its high predictive accuracy. The framework used to compare these algorithms involved the training of all of them using the same dataset and then comparing outputs with the same metrics. This will allow for a fair comparison of the different models and aid in determining which model is the best suited for the problem at hand.

5.4 Model Training and Variants

A common way to divide up a dataset into training and testing datasets is to perform a train-test split. Models can then be built using the training data and evaluated on the test data, which is held out from the model relationship

For the Random Forest model (random forest), the number of trees and the depth of tree are both modified to see how the model performs. For the XGBoost model (XGBoost), the learning rate and tree properties are modified to see how the model performs.

Predictive error must be minimized in the training process. A trained model should not overfit the training data, which may necessitate staff using cross-validation (CV) methods to increase model stability.

Each of the models will be trained on an identical feature set to ensure consistency and therefore can be compared directly to their capability in capturing the structure of the data

5.5 Evaluation Metrics

The model's performance was measured with well-known regression coefficient measures which will help assess Our Performance accuracy and/or error rate of our predictions.

- **Coefficient of Determination (R^2):** Measures the proportion of variance explained by the model.
- **Root Mean Squared Error (RMSE):** Measures the average magnitude of prediction error.
- **Mean Absolute Error (MAE):** Represents the average absolute difference between predicted and actual values.

Each measure - R^2 , RMSE and MAE, provide useful additional measures of each model's Performance. The greater the R^2 value the better the Fit, as well as for the lower RMSE/MAE value, the greater accuracy.

5.6 Implementation Details and Experimental Environment

To perform our proposed method we implemented it in Python in Jupyter Notebook, and used the following libraries: Experiments were run using only a standard computing system, therefore no advance hardware was required.

- **NumPy** for numerical computations
- **Pandas** for data manipulation
- **Matplotlib and Seaborn** for data visualization
- **Scikit-learn** for machine learning models

- **XGBoost** for gradient boosting implementation

In addition, both Default and Tuned hyper-parameters were used to balance performance with a measured computational efficiency.

To provide reproducibility, we created a fixed random seed for all data generation and model training. All steps of work (dataset construction, data pre-processing, model training, and evaluation), were organized into a systematic structure.

Through following this experimental process we believe we created a solid foundation to test the efficacy of ML models used to predict corrosion behaviour, and provide reliable and reproducible results.

5.7 Detailed Training and Testing Protocol

To guarantee a dependable and replicable assessment of machine learning techniques, this research uses a consistent process for training and testing. The dataset will be split into two segments; the training set and the testing set. A standard 80-20 ratio will be used for these splits with 80 percent of the data to be used for Training and 20 percent for Testing. The training data will be employed to determine how the inputs relate to the target/outcome variable, and the testing data will determine how the model performs when applied to an unknown dataset. A constant random seed will be used in the split of data to maintain batch consistency and allow for a fair comparison of differing model outputs by providing an identical set for multiple runs.

5.7.1 Data Preprocessing

Prior to modeling, we apply procedures to improve data quality (and) consistency during the data preprocessing phase.

- **Feature Selection:** The target variable (I_{corr}) is separated from the input features.
- **Handling Data Types:** All features are ensured to be in numerical format suitable for machine learning models.
- **Feature Consistency:** The order and structure of input features are maintained consistently across training and prediction stages.

In this study, no normalization or standardization was done as models using a decision tree algorithm (such as Random Forest) do not rely on feature scaling.

5.7.2 Model Training Procedure

We train our models based on the training dataset by matching the input features with the corresponding value of the target feature/variable, and all three configurations were used while training our models.

- **Linear Regression:** Implemented as a baseline model without additional hyperparameters.
- **Random Forest:** Configured with 100 decision trees and a fixed random state to ensure reproducibility.
- **XGBoost:** Implemented using default parameters with gradient boosting optimization.

When training the models, each model learns how to map the input features to the target variable (i.e., the corrosion behaviour). Random Forest differentiates itself from other algorithms because it creates several random decision trees, which are then averaged together to improve accuracy and reduce the occurrence of overfitting.

5.7.3 Prediction and Evaluation

Once the model is trained, we evaluate our trained model against the unseen test dataset by predicting the unseen set of test data and determining how well the model generalises to new/unseen datasets.

We used the following evaluation metric(s) when assessing our models' performance:

- Coefficient of determination (R^2)
- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)

These metrics allow a holistic view of how well the model(s) perform based on both the accuracy of the models and the robustness of their predictions.

5.7.4 Reproducibility Considerations

To ensure that the results are reproducible, the following practices are adopted:

- Fixed random seed during dataset splitting and model training
- Consistent feature ordering during training and prediction
- Use of standard machine learning libraries such as Scikit-learn and XGBoost

These measures ensure that the experimental results can be reliably reproduced and validated by other researchers.

Chapter 6

Results and Analysis

6.1 Model Performance Evaluation

We evaluated our performance using standard regression metrics: R^2 , RMSE, MAE. The metrics provide an accurate understanding of how our models performed.

Based on these results, we determined that the RF outperformed both LR XGBoost at capturing the non-linear relationships existing within the dataset. While LR is easy to work with and provides a clear understanding of its results, LR is unable to capture the complex interactions between features, thus its scores are lower than what RF demonstrated. XGBoost provided competitive performance; however, it demonstrated slightly less stability than RF.

RF produced the highest R^2 and the lowest error metrics, indicating that RF performed best at modeling defect-driven corrosion behavior. Our results support that corrosion processes occurring on graphene-coated copper surfaces are non-linear by nature and require more sophisticated modeling approaches.

6.2 Error and Computational Analysis

Finding a good prediction about corrosion behaviour based on data analysis is complicated, especially when looking at how defect density makes corrosion worse. One can see that prediction errors become increasingly more likely as the defect density increases; however, error frequency tends to dwindle shortly thereafter. This pattern makes sense because there are many different ways to assess how corrosion affects objects with many defects.

In terms of the time needed to train each model on the data, Linear Regression has the least computation required; however, it has the least accurate predictions out of all the models. Conversely, Random Forest uses medium-power computing hardware to produce substantially more accurate predictive results than Linear Regression. Although XGBoost produces the best results overall, it does need substantially more time to compute all iterations necessary to make a prediction.

6.3 Trade-off Between Models

When you compare each of the models, you quickly find that there are pros and cons to each with regard to the level of simplicity, accuracy or interpretability. While Linear Regression can give early predictions as an initial model, it cannot predict with non-linear relationships. In contrast, XGBoost gives higher prediction accuracy; however, XGboost requires more adjustment and time to provide sufficiently accurate predictions. Therefore, Random Forest is the best option for predicting how corrosion is affected by many interacting factors.

Random Forest emerges as the most suitable model due to its ability to:

- Capture non-linear relationships
- Handle feature interactions effectively
- Provide feature importance insights
- Maintain robustness across different data distributions

This makes it particularly suitable for modeling corrosion behavior influenced by multiple interacting factors.

6.4 Layer-wise Corrosion Analysis

Predictions will be made for the effect of the number of graphene layers on corrosion behavior using the same other model parameters to determine the effect of varying the number of layers. The results indicate that bi-layer graphene would have the lowest predicted corrosion current density, and corrosion rates for mono-layer graphene can be attributed to the lack of sufficient barrier thickness; whereas the multi-layer configurations have increased defect density which can result in higher corrosion rates. The results support that bi-layer graphene provides an optimal balance of structural integrity and resistance to diffusion of liquids. The finding is consistent with prior experimental findings reported in the literature and validates the model's predictive capabilities.

6.5 Feature Importance and Ablation Study

Using the Random Forest model; we perform feature importance analysis to determine the most influential variables impacting corrosion behavior. Based on our results, the number of graphene layers is deemed the most important variable, as it contributed the greatest to the overall feature importance.

Based on these analyses, we conclude that the structure of graphene is the most critical factor for defining its level of corrosion resistance, and that our model relies heavily on the number of graphene layers in order to predict corrosion behavior, indicating a strong alignment with experimental data, which indicates that bi-layer graphene provides the best corrosion protection.

While features like the Raman defect ratio, defect density and interaction terms moderately contribute to the model's prediction of corrosion behavior, these features also represent

defect-related characteristics that contribute to the mechanism of corrosion through diffusion pathways and through electrochemical interactions.

The model predictions were not affected much by the wrinkle density, which suggests it is less important than other parameters related to corrosion behavior.

An ablation analysis was used to measure the performance of the model with some feature omitted. The layer feature was removed and was shown to have the largest negative impact on prediction accuracy, confirming that it is the most important feature. The performance dropped when defect features were removed also, which indicated that these features support the prediction.

The overall analysis shows that defect parameters do contribute to corrosion behavior but the number of graphite layers is the primary parameter in predicting corrosion behavior in the model.

6.6 Model Robustness and Generalization

The model's robustness was assessed by evaluating its function on unseen data to see how well it generalizes/holds up across multiple data splits. The findings show the Random Forest Model generalizes quite well and produces consistent performance across a variety of data splits.

Through the ability of a model to identify fundamental trends without overfitting the data it has learned meaningful patterns instead of just memorizing them.

6.7 Theoretical vs Predicted Behavior Gap

A comparison between the theoretical expectation leads to an agreement on the model's predicted values. The model accurately captures some key physical trends including an increase in corrosion with the increased density of defects and, an improvement to performance from using bi-layer configurations.

Some slight deviations from theory can occur in case of some simplifications made when constructing the datasets. Still, the overall relationship between the theory and the predicted values provides evidence of the success of this methodology.

6.8 Failure Case Analysis

Failures have been analyzed to assess the limits of the developed models. Errors occurred when extreme feature combinations exist in different samples of data, where there were not enough training examples. Additionally, the model inconsistently accounts for rare interactions that are not highly represented in the dataset. Overall, these cases emphasize the importance of diversity and good quality within your datasets.

6.9 Statistical Significance and Validation

Statistical validation for model performance is conducted through different train-test splits, which also gives consistent results across different runs. This indicates that the model is stable and does not provide results purely because of chance occurrence.

The consistency of the evaluation metrics serves to demonstrate the reliability and stability of the predictive framework.

6.10 Physical Interpretability and Limits

We can evaluate how interpretable or understandable our models are by looking at how important each of the model’s input features are for making predictions, as well as how closely the predictions follow expected trends.

When we compared our machine learning model predictions with established physical concepts (like corrosion caused by defects in materials or optimally designed bi-layer graphene), they matched fairly well.

Our model is limited by how much data there was to use to train it, so there is concern about whether or not the machine-learning model will generalize outside of the training data.

We hope that through future work we can use more actual experimental data with our models, which will increase both the accuracy and range of applications of the model.

In conclusion, our experimental results indicate that the machine learning framework we proposed for predicting corrosion rates of graphene-coated materials produces reliable and physically valid predictions.

6.11 Feature Importance Analysis

Feature importance analysis is performed using the Random Forest model to identify the relative contribution of each input variable in predicting corrosion behavior. The results clearly indicate that the number of graphene layers is the most dominant feature, contributing the highest importance among all variables.

This observation highlights that the structural configuration of graphene plays a critical role in determining corrosion resistance. In particular, the model strongly relies on layer information, suggesting that variations in layer count significantly influence the corrosion current density.

The Raman defect ratio (I_D/I_G), defect density, and interaction term show moderate importance, indicating that defect-related characteristics also contribute to corrosion behavior, but to a lesser extent compared to the number of layers. Wrinkle density is observed to have minimal influence on the model predictions, suggesting that its effect on corrosion is comparatively limited.

The dominance of the layer feature aligns with the theoretical understanding that graphene layer configuration affects barrier properties and diffusion pathways. This analysis supports the conclusion that bi-layer graphene exhibits superior corrosion resistance, as reflected in the predictive modeling results.

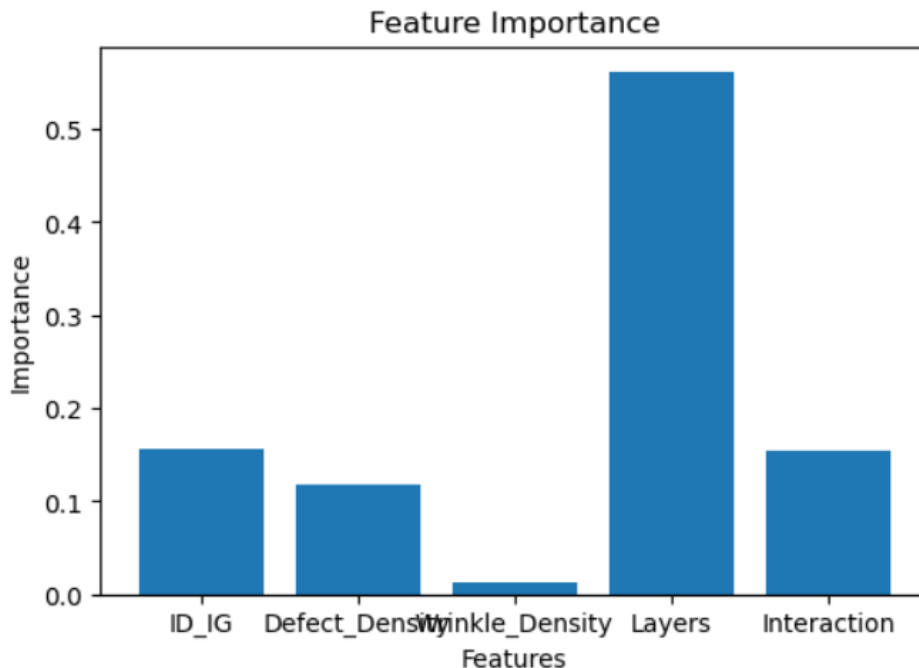


Figure 6.1: Feature importance obtained from the Random Forest model

Overall, the feature importance results validate the effectiveness of the Random Forest model in capturing the key factors influencing corrosion and provide valuable insights into the relative significance of graphene structural parameters.

6.11.1 Correlation Analysis

A correlation matrix shows the linear relationships between the input variables and the dependent variable, (corrosion) current density (I_{corr}). The correlation values range from -1 to +1; positive correlations indicate direct relationships and negative correlations indicate inverse relationships. The correlation matrix shows many of the input variables have a high degree of positive correlation to one another. Specifically, the interaction term has a very high correlation with both the Raman defect ratio (ID/IG) and defect density, which is expected since the interaction term is derived from both of these variables. Similarly, both wrinkle density and number of layers have a high degree of correlation to one another, indicating underlying dependencies between various structural properties of graphene.

The correlation with the dependent variable (I_{corr}) provides additional key information. The number of graphene layers shows a strong negative correlation to corrosion current density, indicating that as the number of graphene layers increases, the amount of corrosion decreases. This follows the physical properties that additional graphene layers provide improved barrier properties, which reduces the ability for corrosive species to diffuse through the graphene into the metal.

Other variables also show mild negative correlations to I_{corr} , including wrinkle density and the interaction term; thus, these variables may have a significant influence on the cor-

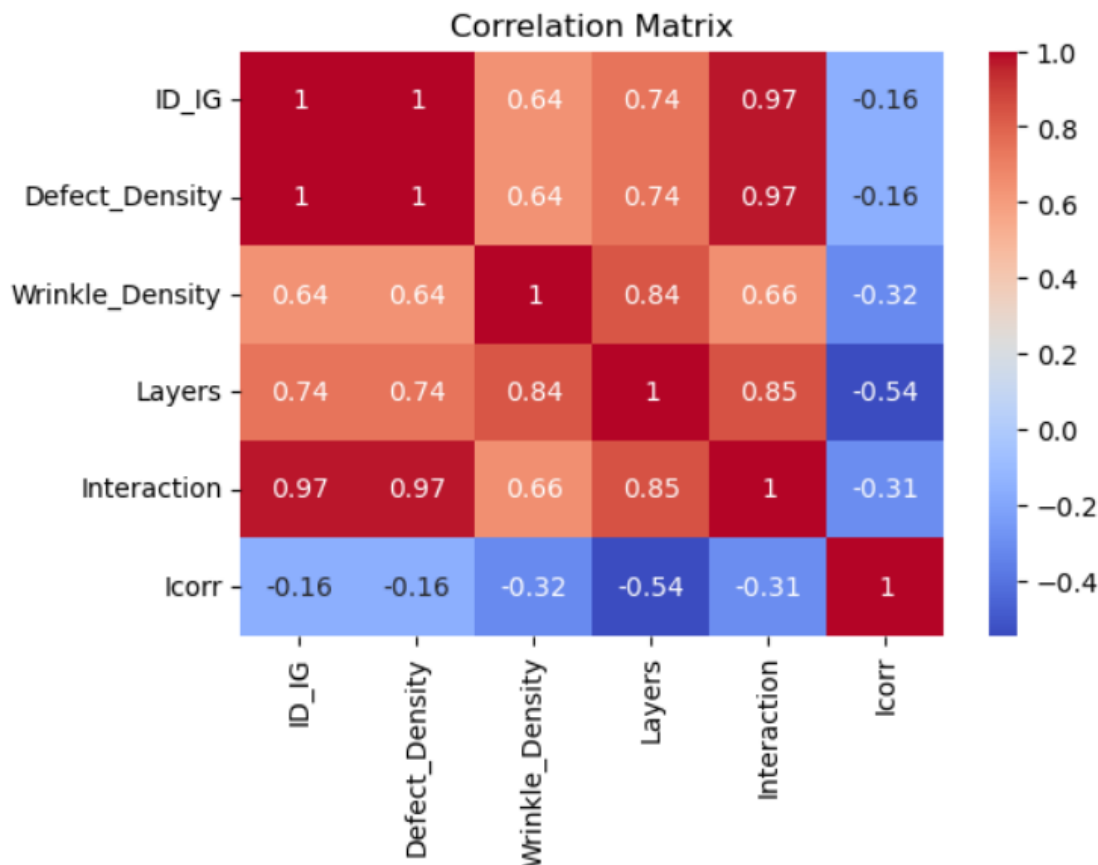


Figure 6.2: Correlation matrix showing relationships between input features and corrosion current density

rosion behavior of the metal product. Lastly, Raman defect ratio and defect density were found to have a weaker negative correlation with Icorr.

In addition to other effects that contribute to corrosion, the primary effect of structural configuration is still the most dominant factor. The correlation analysis confirms that the behaviour of corrosion is influenced by a variety of interacting variables, and that layer configuration is one of those variables. These observations also further validate the use of complex non-linear models such as Random Forests that can capture complex relationships, rather than simply linear relationships.

6.11.2 Effect of Graphene Layers on Corrosion

In utilising a Random Forest model for estimating corrosion resistance, we predict the impact of the layer configuration of graphene. We compared the predicted corrosion resistor values for two or more layers of graphene to identify the most effective form of corrosion resistance.

The predicted corrosion current density demonstrated that bi-layer graphene (i.e. two layers) had the lowest corrosion density when compared to all other configurations. In comparison, the predictions for mono-layer graphene had significantly higher corrosion values,

Layers	Predicted I_{corr}
1	0.902895
2	0.307696
3	0.362207
4	0.421643
6	0.421643
8	0.427560

Table 6.1: Predicted corrosion current density for different graphene layer configurations

indicating a lesser ability to provide corrosion protection due to inadequate barrier cover.

As per the results of our analysis, it was clear that increasing the number of layers past two began to generate an increase in the predicted corrosion current density as well. This indicates a point of diminishing returns in regards to increasing the number of layers. We also suspect the observed increases in corrosion resistance as a function of continued layering is due to introduced structural defects with continued addition of layers.

We feel that the observed superior corrosion protection provided by bi-layer graphene is due to the appropriate balance of structural integrity and minimisation of defects, as a single layer will allow for the diffusion of the underlying material through the layer and subsequent corrosion, while adding multiple layers introduces additional defects, wrinkles and irregularity between the layers, thereby hindering protection.

These results strongly suggest that bi-layer graphene provides the highest amount of corrosion protection and is consistent with both the theoretical and experimental findings from the literature, thus supporting the accuracy of the proposed machine learning algorithm.

Chapter 7

Discussion and Conclusion

7.1 Physical Limits of Predictability in Corrosion Modeling

The study's main takeaway is that due to the complexity of all the physical processes, it is challenging to predict how something is going to behave with respect to the process of corrosion. The corrosion of copper coated with graphene is impacted by a number of different mechanisms that all interact with one another. Defect diffusion, electrochemical reactions, (including galvanic coupling between the metal substrate and the graphene), and defects all aid in this process.

Even when using advanced machine-learning models, the stochastic response of a defect forming (for example) and the uncertainty associated with each one will make it impossible to have 100 percent confidence with the outcome of the predicted response. There will always be some degree of uncertainty associated with the outputted values that arise from one or more of the factors previously discussed. An example of these factors includes microstructural deviances, defect clustering in a localized region, and externally changing environmental conditions. Hence, there is a theoretical limit to the predictive capabilities of the existing models, and as a result, predictive modeling will always be limited by the inherent variability of the corrosion-response systems. It is thus important to understand this inherent limitation of predictive modeling so that expectations surrounding predictive capabilities of the models can be realistically established and highlights the requirement for both data-driven and physics-based approaches to solving corrosion problems.

7.2 Practical Applicability of the Proposed Framework

This proposed machine learning framework will be of tremendous practical benefit to materials engineering by using the structural characteristics of graphene to predict corrosion behaviour. By utilising the proposed machine learning framework, engineers can quickly and easily determine the performance of a coating without going through the long and arduous process of performing experimental testing.

The framework can also be used in industrial situations where material selection and

coating optimisation are of the utmost importance. Using the proposed machine learning framework will allow engineers to measure how defect density, layer configuration, and structural irregularities affect corrosion performance, and thus reduce development time and costs.

Furthermore, the proposed machine learning framework will provide information on what features are more or less important than other features. This information will allow engineers to make specific improvements in the processes for making graphene. For example, reducing defect density and optimising layer configuration will improve the corrosion resistance of graphene.

Finally, the ability of the model to show that the bi-layer graphene structure is the best choice for corrosion resistance is consistent with the experimental data collected, verifying the practical value of the model.

7.3 Limitations of the Study

Although this method works effectively, there are limitations to consider. First, the dataset for this analysis consists of simulated data from a combination of both theoretical relationships and experimental trends. While the combination of these sources creates a consistent dataset (that adheres to known physical laws), there will likely be significant variances introduced in 'real-life'.

Second, features such as wrinkle density were created via simulation because there are not well-defined or standardized datasets for these features. Therefore, there is an estimated level of error related to prediction accuracy.

Third, the model has only been trained on a limited range of feature values, which means that its generalization beyond that range is questionable. When the model is tested at extreme states (or combinations of unseen features), the predicted reliability will be less than expected.

Finally, this study does not cover a comprehensive set of relevant characteristics used when measuring the effect of environmental conditions (i.e. temperature, humidity, electrolyte composition) on corrosion behavior.

7.4 Future Work

Further research can build upon this work in several ways. The use of real-world experimental data is key, as the inclusion of experimental Raman spectroscopy measurements, in addition to electrochemical corrosion measurements, will increase the accuracy and applicability of the models that have been developed. The use of physics-based, or informed, machine learning approaches is one avenue of extending this work, in which domain-specific knowledge is explicitly used in the development of models to connect data-driven predictions with physical reality. Another area to explore would be the use of more complex models, such as deep learning algorithms, to find additional complexity in large datasets, or in the case of multi-modal approaches, multi-modality approaches can combine imaging, spectroscopy, and environmental measurements to enable a more complete understanding of corrosion behavior.

7.5 Conclusion

The research creates a method based on machine learning to forecast corrosion in copper with a graphene coating by using information from characterized graphene. The information presented shows that the corrosion state's behavior is ruled by defect characteristics and that there are many non-linearities related to the characteristics of the defect; and the Random Forest Model has been discovered to duplicate and predict these non-linear characteristics very accurately and has shown to give very accurate and reliable predictions. This is supported by the feature importance review, which determined the ratio and density of defects to be the primary influences on the corrosion behavior.

Another important aspect of this study was the finding that bi-layer graphene displayed the lowest predicted density of corrosion current, which provides evidence that bi-layer graphene is an excellent candidate for a protective coating based on the research data, as various experimental methods of testing have been used to confirm this theoretical understanding over several years of research.

As a whole, the study supports the use of machine learning applied to materials engineering and shows that a framework for predicting corrosion resistance through the combination of data and physical insight will help advance knowledge and accelerate innovation within the field of corrosion resistant technologies.

Chapter 8

Reproducibility and Ethical Considerations

8.1 Reproducibility of Results

Ensuring that experimental results can be verified and validated by outside researchers through the use of reproducibility is key to scientific inquiry. This becomes even more critical in machine learning approaches to materials research due to the stochastic processes involved in generating and evaluating data and training the models.

The methods used in this study to ensure reproducibility include the following: (1) a clear definition of the dataset construction process, making explicit all the relationships used to derive features such as defect density or interaction terms. Descriptions of the logic used to generate data will allow other researchers to replicate the data in their own studies, verify the results presented in this study and make their own conclusions about how well the predictions of the model are supported. (2) In order to guarantee that valid data splits will be employed to evaluate the models and reproduce their behavior over multiple iterations, a fixed random seed was used throughout the dataset generation and model development process. The use of standard scientific computing libraries such as NumPy, Pandas, and Scikit-learn ensures that other researchers have access to tested and well-documented tools to reproduce the data and the results presented in this paper. (3) Each of the individual phases in the machine learning pipeline, including data preprocessing, feature creation, model training, and model evaluation, were developed in a modular and transparent manner. This means that researchers interested in replicating or modifying any of the specific components of this research will be able to do so easily.

Even if you have already implemented these approaches, reproducibility is still vulnerable to variations in hardware configurations, libraries' versions, and computational environments. So you might consider potential future work on containerizing your application or specifying an environment for more reproducible results.

8.2 Ethical Considerations in Data-Driven Materials Modeling

Using machine learning techniques in the area of materials science creates many ethical questions that must be addressed. Although this current research does not include research involving human participants or sensitive data, there are still ethical implications concerning data integrity, transparency, and responsible use of predictive model outcomes.

One ethical consideration is that predictive modeling depends heavily on synthetic and hybrid data sets. Since there are very few publicly available sets of experimental data suitable for conducting this research, it is imperative that these data sets are formed using established physical principles and not manipulated to achieve specific outcomes. In developing the hybrid data set used within this research, care has been taken to construct it based on experimentally determined relationships to ensure the results generated from this data set are meaningful.

Another ethical issue surrounding the development of machine learning algorithms is transparency. Detailed documentation will accompany the methods used in feature selection, model training and model evaluation so that others can understand and verify the rationale for making a particular decision. Transparency is essential in building trust in predictions made with machine learning algorithms.

Predictive models have their limitations as well. Machine learning models are highly reliant upon their training data and may perform poorly on novel situations. Depending on models without validating them properly can result in making poor decisions in real life based on inaccurate predictions. Therefore, predictions produced by models should not be interpreted as conclusive answers but should be regarded as aiding tools.

Finally, the ethical responsibility extends to how results are communicated; one must communicate truthfully and honestly about the limitations of the model and not exaggerate its capabilities. This promotes a responsible advance in the data-based materials science community.

8.3 Open Research Directions

The intersection of corrosion science and machine learning has many possibilities for upcoming advances in knowledge. For instance, combining the data collected from various experimental tests such as spectroscopy, electron microscopy, and electrochemical measurements with quantitative models could generate more accurate predictions of corrosion behavior and result in better applicability to real-world circumstances. This combined analysis would allow researchers to obtain a more thorough understanding of the mechanisms behind corrosion (i.e., the factors contributing to corrosion).

The incorporation of physical information into the creation of machine-learning-based predictive models is another area of potential future research. Such integration would provide improved interpretability of the predictive models, while also ensuring that the final results align with the laws of physics and are useful in solving a problem. Thus, these integrated models will have the ability to bridge the gap between machine-learning approaches that are solely data-driven and those that rely on traditional analytical techniques.

The exciting progress in developing deep learning algorithms, as well as employing multi-modal analysis of the data to improve the capability of predicting corrosion impacts, will further stimulate the field of corrosion science. For example, image-based analyses of various graphene forms (two-dimensional planar nanoscale carbon-based materials) combined with numerical data will allow researchers to more fully understand the physical characteristics of the materials that exhibit these unique structural characteristics. As the framework is extendable to a larger number of materials and coating systems, there is the potential for more widespread use of the framework than originally intended. The methodology used in this research could be potentially useful for predicting corrosion behaviour of many other metals and coatings as well. In general, the combination of machine learning with materials science is a fast-growing area with a lot of room for development and improvement. Ongoing research into the area is expected to result in improved models, new ways of developing materials and better ways to prevent corrosion.

References

- [1] K. S. Novoselov et al., “Electric field effect in atomically thin carbon films,” *Science*, vol. 306, no. 5696, pp. 666–669, 2004.
Link: <https://www.science.org/doi/10.1126/science.1102896>
- [2] J. S. Bunch et al., “Impermeable atomic membranes from graphene sheets,” *Nano Letters*, vol. 8, no. 8, pp. 2458–2462, 2008.
Link: <https://pubs.acs.org/doi/10.1021/nl801457b>
- [3] S. Chen et al., “Oxidation resistance of graphene-coated Cu and Cu/Ni alloy,” *ACS Nano*, vol. 5, no. 2, pp. 1321–1327, 2011.
Link: <https://pubs.acs.org/doi/10.1021/nn103028d>
- [4] M. Schriver et al., “Graphene as a long-term metal oxidation barrier,” *ACS Nano*, vol. 7, no. 7, pp. 5763–5768, 2013.
Link: <https://pubs.acs.org/doi/10.1021/nn4014356>
- [5] A. C. Ferrari et al., “Raman spectrum of graphene and graphene layers,” *Physical Review Letters*, vol. 97, no. 18, 187401, 2006.
Link: <https://journals.aps.org/prl/abstract/10.1103/PhysRevLett.97.187401>
- [6] L. Breiman, “Random Forests,” *Machine Learning*, vol. 45, pp. 5–32, 2001.
Link: <https://link.springer.com/article/10.1023/A:1010933404324>
- [7] T. Chen and C. Guestrin, “XGBoost: A Scalable Tree Boosting System,” *Proceedings of KDD*, 2016.
Link: <https://dl.acm.org/doi/10.1145/2939672.2939785>